Chapter 13: Standardization and the Parametric G-Formula

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6 Summary 11

This chapter estimates the same causal effect as Chapter 12 (smoking cessation on weight gain in NHEFS) by **standardization**, now combined with outcome models: the **parametric g-formula**. IP weighting models the treatment (and censoring); standardization models the outcome. Both rely on the same identifiability conditions but on different modeling assumptions.

This chapter is based on Hernán and Robins (2020, chap. 13, pp. 175-185).

R and Stata code for this chapter's programs (Programs 13.x) is in Tom Palmer's cibookex-r[1] companion (GPL-3.0), which descends from the authors' own code.

Chapter 2 introduced standardization only as a nonparametric method. Combining it with models lets us handle many covariates and nondichotomous treatments.

Key insight: IP weighting needs a correct model for treatment (and censoring) given L; the parametric g-formula needs a correct model for the outcome given treatment and L. Neither is doubly robust on its own. Doubly robust estimators (Section 13.4) combine the two models and are consistent if *either* one is correct.

1 13.1 Standardization as an Alternative to IP Weighting (pp. 175-177)

[1] https://remlapmot.github.io/cibookex-r/

Remark 1.1 (NHEFS: Setting and Target Effect). **Setting** (same as Chapter 12):

- NHEFS: 1629 cigarette smokers aged 25-74 with a baseline visit and a follow-up visit about 10 years later
- 1566 had weight measured at follow-up, i.e., are uncensored ($C = 0$)
- Treatment A: smoking cessation (1: yes, 0: no); outcome Y: weight gain (kg)

Causal effect of interest:

$$\mathrm{E}[Y^{a=1,c=0}] - \mathrm{E}[Y^{a=0,c=0}]$$

the difference in mean weight gain had everybody been treated and uncensored versus untreated and uncensored.

Quitters and non-quitters differ in the distribution of predictors of weight gain (Table 12.1), so the associational difference $\mathrm{E}[Y \mid A = 1, C = 0] - \mathrm{E}[Y \mid A = 0, C = 0] = 2.5$ kg is not expected to equal the causal difference.

As in Chapter 12, we assume that the 9 baseline variables in L suffice to adjust for both confounding and selection bias:

- sex (0: male, 1: female), age (years), race (0: white, 1: other)
- education (5 categories)
- intensity and duration of smoking (cigarettes per day; years of smoking)
- physical activity in daily life (3 categories), recreational exercise (3 categories)
- weight (kg)

We also assume, as in Chapter 12, that the components of L needed to adjust for C are not affected by A; otherwise the methods of Part III would be needed.

IP weighting estimates $\mathrm{E}[Y^{a,c=0}]$ as the mean outcome in a pseudo-population where L is balanced between treatment groups; it requires modeling the joint probability $\Pr[A = a, C = 0 \mid L]$.

1.1 The Standardized Mean

Definition 1.1 (Standardized mean)**.** The **standardized mean** outcome in the uncensored who received treatment level a is

$$\sum_{l} \mathrm{E}[Y \mid A = a, C = 0, L = l] \times \Pr[L = l],$$

the average of the stratum-specific means, weighted by the prevalence of each stratum l in the study population. When some components of L are continuous, $\Pr[L = l]$ is replaced by the density $f_L(l)$ and the sum becomes an integral, $\int \mathrm{E}[Y \mid A = a, C = 0, L = l]\, dF_L(l)$, where F_L is the joint cdf of L.

Theorem 1.1 (Standardization identifies the counterfactual mean)**.** *Let L be discrete, and fix a treatment level a. Assume*

- *conditional exchangeability for the joint treatment (A, C): $Y^{a,c=0} \perp\!\!\!\perp (A, C) \mid L$;*
- *positivity: $\Pr[A = a, C = 0 \mid L = l] > 0$ for every l with $\Pr[L = l] > 0$;*
- *consistency: $Y^{a,c=0} = Y$ for every individual with $A = a$ and $C = 0$.*

Then the standardized mean of Definition 1.1 equals $\mathrm{E}[Y^{a,c=0}]$.

Proof

Proof. For discrete L (Technical Point 2.3 of the book, with $A = a$ replaced by $(A = a, C = 0)$):

$$
\begin{aligned}
\mathrm{E}[Y^{a,c=0}] &= \sum_l \mathrm{E}[Y^{a,c=0} \mid L=l]\Pr[L=l] && \text{(law of total expectation)}\\
&= \sum_l \mathrm{E}[Y^{a,c=0} \mid A=a, C=0, L=l]\Pr[L=l] && \text{(exchangeability for } (A,C) \text{ given } L\text{; positivity)}\\
&= \sum_l \mathrm{E}[Y \mid A=a, C=0, L=l]\Pr[L=l] && \text{(consistency)}
\end{aligned}
$$

Positivity is needed in the second line so that the conditional mean given $(A=a, C=0, L=l)$ is defined for every l with $\Pr[L=l]>0$. □

The same argument, with sums replaced by integrals, covers continuous components of L. So, under these conditions, any estimator that converges to the standardized mean with $a=1$ also converges to $\mathrm{E}[Y^{a=1,c=0}]$; likewise with $a=0$ and $\mathrm{E}[Y^{a=0,c=0}]$.

1.2 Supplement: Standardizing to the Treated

Remark 1.2 (Standardizing to the treated). The average causal effect **in the treated** can also be estimated by standardization (book's Technical Point 4.1): replace $\Pr[L=l]$ by $\Pr[L=l \mid A=1]$ in Definition 1.1.

More generally, the stratum-specific means can be standardized to any distribution of L (the treated, the untreated, an external population). The choice defines the target population of the causal effect.

ℹ Fine Point 13.1: Structural Positivity

- Positivity is needed for standardization too: if $\Pr[A=a, C=0 \mid L=l]=0$ while $\Pr[L=l]\neq 0$, then $\mathrm{E}[Y \mid A=a, C=0, L=l]$ is undefined.
- With a parametric outcome model, one *can* ignore structural nonpositivity by **extrapolating** over the empty strata, at the cost of bias (95% confidence intervals then cover the truth less than 95% of the time).
- Near positivity violations, standardization typically has a smaller standard error than IP weighting, but differences in bias may outweigh differences in standard error.

Hernán and Robins (2020, Fine Point 13.1, p. 176) stresses the different role of modeling here: we model not because data are scarce but to estimate a quantity that is not identified even with infinite data, because of structural nonpositivity.

2 13.2 Estimating the Mean Outcome via Modeling (pp. 177-178)

Remark 2.1 (Why we model the outcome). **Problem**: nonparametric estimation of $\mathrm{E}[Y \mid A=1, C=0, L=l]$ (as in Section 2.3) is out of the question. Only 403 uncensored treated individuals are spread across possibly millions of strata of the 9 confounders. Same for the untreated.
Solution: fit a parametric model for the mean outcome.

Example 2.1 (NHEFS outcome model)**.** A linear regression model for $\mathrm{E}[Y \mid A, C=0, L]$ (among the uncensored) with covariates:

- treatment A and all 9 confounders in L
- linear and quadratic terms for age, weight, smoking intensity, and smoking duration
- a product term $A\times$ smoking intensity

The model assumes each covariate contributes additively to the mean, except that the contribution of A varies linearly with smoking intensity.

Fitting this model (book's Program 13.1) gives a predicted mean $\hat{\mathrm{E}}[Y \mid A = a, C = 0, L = l]$ for each combination of A and L, and hence for each of the 403 uncensored treated and 1163 uncensored untreated individuals.

For example, a non-quitter who is male, white, age 26, a college dropout, smoking 15 cigarettes/day for 12 years, with moderate exercise, very active, and weighing 112 kg had a predicted mean weight gain of 0.34 kg (individual 24770 in the data).

The mean of the predicted values was 2.6 kg, matching the mean observed weight gain (observed values ranged from −41.3 to 48.5 kg).

2.1 Supplement: Other Outcome and Treatment Types

Remark 2.2 (Other outcome and treatment types). The same approach applies with other outcome models:

- **Dichotomous** Y: fit, e.g., a logistic model for $\Pr[Y = 1 \mid A, C = 0, L]$; the standardized means are then standardized risks, from which a risk difference, risk ratio, or odds ratio can be computed.
- **Nondichotomous** A: standardize the predicted means at each treatment level a of interest.

⚠ Conditional Odds Ratios Are Not Marginal

The conditional (on L) odds ratio from a logistic outcome model is generally not equal to the marginal causal odds ratio, because the odds ratio is not collapsible; standardization averages predicted *risks* first and then forms the effect measure.

3 13.3 Standardizing the Mean Outcome to the Confounder Distribution (pp. 178-179)

We do **not** need to estimate $\Pr[L = l]$.

Proposition 3.1 (The standardized mean is an iterated expectation). *For discrete L,*

$$\sum_{l} \mathrm{E}[Y \mid A = a, C = 0, L = l] \Pr[L = l] = \mathrm{E}[\mathrm{E}[Y \mid A = a, C = 0, L]]$$

ℹ Proof

Proof. The outer expectation is over the marginal distribution of L. Writing $g(l) = \mathrm{E}[Y \mid A = a, C = 0, L = l]$, we have $\mathrm{E}[g(L)] = \sum_l g(l) \Pr[L = l]$ by the definition of expectation. □

By Proposition 3.1, the standardized mean can be estimated by averaging the model predictions over the **empirical distribution** of L:

$$\frac{1}{n} \sum_{i=1}^{n} \hat{\mathrm{E}}[Y \mid A = a, C = 0, L = L_i]$$

where n is the number of individuals in the study.

Replacing g by its estimate $\hat{g}$ and the distribution of L by the empirical distribution (mass $1/n$ on each observed L_i) gives $\frac{1}{n}\sum_i \hat{g}(L_i)$.

Averaging over the observed L_i is a nonparametric estimate of the distribution of L, which is valid here because L consists of baseline covariates unaffected by treatment.

3.1 The Four-Step Algorithm

Definition 3.1 (Standardization by expanding the dataset)**.**

1. **Expansion**: make three copies of the data. Block 1 is the original data. In block 2 set $A = 0$ for everyone; in block 3 set $A = 1$ for everyone. Set Y to missing in blocks 2 and 3.
2. **Outcome modeling**: fit the outcome model on the 3-block dataset. Only block 1 contributes, because Y is missing elsewhere.
3. **Prediction**: use the fitted model to predict Y for every row of blocks 2 and 3.
4. **Standardization by averaging**: the mean prediction in block 2 estimates the standardized mean under $A = 0$; in block 3, under $A = 1$.

3.2 Example: Table 2.2 (No Censoring, One Binary L)

Example 3.1 (Standardization of the Table 2.2 data)**.** The 20-row Table 2.2 becomes 60 rows. With a saturated model (main effects of A and L plus $A \times L$), the predictions equal the stratum-specific risks: $\Pr[Y = 1 \mid A = a, L = 0] = 1/4$ and $\Pr[Y = 1 \mid A = a, L = 1] = 2/3$ for both $a = 0$ and $a = 1$. Since 60% of rows have $L = 1$ and 40% have $L = 0$, each block's average prediction is

$$\frac{1}{4} \times 0.4 + \frac{2}{3} \times 0.6 = 0.1 + 0.4 = 0.5,$$

exactly the standardized risks of Section 2.3 (0.5 under both $a = 0$ and $a = 1$).

With a saturated model, this procedure is completely nonparametric, so it reproduces the direct calculation of Chapter 2 (book's Program 13.2). The averaging step weights each stratum by its share of rows, which is how it implements $\Pr[L = l]$ without estimating it.

3.3 Example: NHEFS

Example 3.2 (Parametric g-formula in NHEFS)**.** Applying the four steps with the outcome model of Example 2.1 (book's Program 13.3):

- standardized mean in the untreated (block 2): 1.66 kg
- standardized mean in the treated (block 3): 5.18 kg
- estimated effect: $5.18 - 1.66 = 3.52 \approx 3.5$ kg

With a nonparametric bootstrap (Program 13.4), the 95% confidence interval is 2.6 to 4.5 kg.

⚠ Two Different Confidence Intervals

The two intervals in the book belong to different methods. The bootstrap interval (2.6, 4.5) is the parametric g-formula's own (Hernán and Robins 2020, chap. 13, p. 179). The interval (2.5, 4.5) is the one for the IP weighted estimate with weights $SW^{A,C}$ in Chapter 12 (Hernán and Robins 2020, chap. 12, p. 174). The summary in Section 13.5 states one interval, (2.5, 4.5), for both methods together (Hernán and Robins 2020, chap. 13, p. 181); that interval matches the IP weighted one, and the g-formula's own is (2.6, 4.5).

Technical Point 13.1: Bootstrapping

The **nonparametric bootstrap** estimates standard errors when analytic or large-sample formulas are impractical:

1. Draw a sample of size 1629 **with replacement** from the 1629 individuals (a bootstrap sample).
2. Compute the effect estimate in the bootstrap sample with the same method (here, standardization).
3. Repeat many times (the book uses 1000).
4. The standard deviation of the bootstrap estimates estimates the standard error; the 95% CI is the estimate $\pm 1.96\times$ that standard error.

- When an estimator's limiting distribution is normal, Wald intervals with bootstrap standard errors are calibrated in large samples. For IP weighted MSM estimates, the bootstrap interval is calibrated, whereas the robust ("sandwich") variance from Chapter 12 often gives a conservative, wider interval.
- Published analyses often use 200-500 bootstrap samples to save computation; in this example that would have given an almost identical interval, but not necessarily in general. The "bag of little bootstraps" (Kleiner et al. 2014) is a scalable alternative.
- Wasserman (2004) introduces the theory behind the bootstrap.

4 13.4 IP Weighting or Standardization? (pp. 179-180)

Remark 4.1 (IP weighting and standardization model different things).

- Without models, the IP weighted and standardized means are **exactly equal** (Technical Point 2.3).
- With models, they generally **differ**, because they model different things:
  - IP weighting: $\Pr[A = a, C = 0 \mid L]$ (in Chapter 12, logistic models for $\Pr[A = a \mid L]$ and $\Pr[C = 0 \mid A = a, L]$)
  - Standardization: $\mathrm{E}[Y \mid A = a, C = 0, L]$ (here, a linear model)
- Some misspecification is inescapable, and it will usually bias the two estimates differently.

Compare the Two Estimates

- A **large** difference between the two estimates signals serious misspecification in at least one of them.
- A **small** difference does not guarantee correct models, but it is reassuring: badly misspecified models are unlikely to produce biases of similar magnitude and direction.
- In NHEFS both estimates round to **3.5 kg**.
- Neither method modeled the confounders L: IP weighting does not need their distribution, and standardization used their empirical distribution.

4.1 The (Parametric) G-Formula

Definition 4.1 (Parametric g-formula)**.** IP weighting and standardization each estimate the **g-formula** (Robins 1986). Standardization is a **plug-in** g-formula estimator: it substitutes an estimate for the conditional mean outcome that appears in the g-formula. When that estimate comes from a parametric model, the method is the **parametric g-formula**.

Remark 4.2 (What the parametric g-formula needs to model).

- For an average causal effect we only need the conditional **mean** outcome; for a standardized **distribution** (pdf) we would need the conditional distribution of Y given A and L.
- Without time-varying confounders, the parametric g-formula does not require modeling the distribution of L.

Terminology: G-Formula, G-Computation, G-Estimation

Robins (1986) generalized standardization to time-varying treatments and confounders under the name *g-computation algorithm formula*. Some authors shorten this to "g-formula", others to "g-computation". The book uses only "g-formula", because "g-computation" is often confused with g-estimation, a different method (Chapter 14).
Part III defines the g-formula for time-varying treatments.

4.2 Use Both, and Use Doubly Robust Methods

Definition 4.2 (Doubly robust estimator)**.** An estimator of a counterfactual mean such as $\mathrm{E}[Y^{a=1}]$ that combines a treatment model and an outcome model is **doubly robust** if, under the identifying conditions (conditional exchangeability and positivity given L, and consistency), it is consistent for that counterfactual mean when **either** model is correctly specified, without our knowing which. (With censoring, replace Y^a by $Y^{a,c=0}$ and A by the joint treatment (A, C) throughout.)

An estimator of a causal contrast, such as $\mathrm{E}[Y^{a=1}] - \mathrm{E}[Y^{a=0}]$, built from doubly robust estimators (Definition 4.2) of each counterfactual mean is then consistent for that contrast, provided the identifying conditions hold for every treatment level in the contrast (here both $a = 1$ and $a = 0$) and, for each counterfactual mean, at least one of its two models is correctly specified.

Use Both Methods, and Doubly Robust Ones

- When both IP weighting and the parametric g-formula can be used, **use both**.
- Whenever possible, use **doubly robust** estimators (Definition 4.2).
- Both methods can also target a subset of the population, e.g., estimating standardized means separately in men and women to study effect modification by sex.

Fine Point 13.2: A Doubly Robust Plug-In Estimator

For dichotomous A and no censoring (Bang and Robins 2005):

1. Estimate the IP weights $W^A = 1/f(A \mid L)$ as in Chapter 12.
2. Fit an outcome model (a GLM with canonical link) for $\mathrm{E}[Y \mid A, L, R]$ that adds the covariate $R = W^A$ if $A = 1$ and $R = -W^A$ if $A = 0$.
3. Predict for everyone with A set to 1 and R recomputed at that value, $R = 1/f(1 \mid L)$, and average; repeat with A set to 0 and $R = -1/f(0 \mid L)$.
4. The difference of the two averages is a doubly robust plug-in estimate of the average causal effect.

Equivalently, $R = \{I(A = 1) - I(A = 0)\}/f(A \mid L)$, which is the "clever covariate" of targeted minimum loss-based estimation (TMLE); see the book's Technical Point 13.3. Including R makes the fitted model's residuals, weighted by R, sum to zero, which is what makes the plug-in estimator doubly robust.

Definition 4.3 (Augmented IP weighted (AIPW) estimator)**.** Let A be dichotomous, with no censoring, and define

- the outcome regression $b(L) \stackrel{\text{def}}{=} \mathrm{E}[Y \mid A = 1, L]$;
- the propensity score $\pi(L) \stackrel{\text{def}}{=} \Pr[A = 1 \mid L]$.

Let $\hat{b}(L)$ be the predicted mean outcome under $A = 1$ from a fitted model for $\mathrm{E}[Y \mid A, L]$ (the outcome model), and let $\hat{\pi}(L)$ be the predicted probability of $A = 1$ from a fitted model for $\Pr[A = 1 \mid L]$ (the treatment model). The **AIPW estimator** of $\mathrm{E}[Y^{a=1}]$ is

$$\hat{\mathrm{E}}[Y^{a=1}]_{DR} = \frac{1}{n}\sum_{i=1}^{n}\left[\hat{b}(L_i) + \frac{A_i}{\hat{\pi}(L_i)}\left\{Y_i - \hat{b}(L_i)\right\}\right]$$

Technical Point 13.2: The Augmented IP Weighted Estimator

For dichotomous A and no censoring, with b, π, $\hat{b}$ and $\hat{\pi}$ as in Definition 4.3, under the identifiability conditions,

$$\mathrm{E}[Y^{a=1}] = \mathrm{E}[b(L)] = \mathrm{E}\left[\frac{AY}{\pi(L)}\right].$$

- Plug-in g-formula estimator: $\frac{1}{n}\sum_i \hat{b}(L_i)$
- Horvitz-Thompson IP weighted estimator: $\frac{1}{n}\sum_i A_i Y_i/\hat{\pi}(L_i)$

Two equivalent forms. Expanding the summand of the AIPW estimator (Definition 4.3):

$$\begin{aligned}\hat{b}(L_i) + \frac{A_i}{\hat{\pi}(L_i)}\{Y_i - \hat{b}(L_i)\} &= \hat{b}(L_i) + \frac{A_i Y_i}{\hat{\pi}(L_i)} - \frac{A_i}{\hat{\pi}(L_i)}\hat{b}(L_i) \quad \text{(distribute)}\\ &= \frac{A_i Y_i}{\hat{\pi}(L_i)} - \left(\frac{A_i}{\hat{\pi}(L_i)} - 1\right)\hat{b}(L_i) \quad \text{(collect the } \hat{b}(L_i) \text{ terms)}\end{aligned}$$

The first form is the outcome-model estimator plus an IP weighted correction; the second is the Horvitz-Thompson estimator plus an outcome-model correction ("augmentation"), hence the name.

4.3 Why the AIPW Estimator Is Doubly Robust

Theorem 4.1 (Double robustness of the AIPW functional)**.** *Let A be dichotomous, with no censoring, and let b and π be as in Definition 4.3. Assume*

- *conditional exchangeability: $Y^{a=1} \perp\!\!\!\perp A \mid L$;*
- *positivity: $\pi(l) > 0$ for every l in the support of L;*
- *consistency: $Y^{a=1} = Y$ for every individual with $A = 1$;*
- *$\mathrm{E}[|Y|] < \infty$ and $\mathrm{E}[|Y^{a=1}|] < \infty$.*

Let b^* and π^* be any functions of L with $\mathrm{E}[|b^*(L)|] < \infty$ and $\pi^*(l) \geq \epsilon$ for some $\epsilon > 0$ and every l in the support of L. Then

$$\mathrm{E}\left[b^*(L) + \frac{A\{Y - b^*(L)\}}{\pi^*(L)}\right] - \mathrm{E}[Y^{a=1}] = \mathrm{E}\left[\pi(L)\left(\frac{1}{\pi^*(L)} - \frac{1}{\pi(L)}\right)(b(L) - b^*(L))\right],$$

*which is zero if $\pi^* = \pi$ **or** $b^* = b$.*

Proof

Proof. First condition on L:

$$\mathrm{E}\left[\frac{A\{Y-b^*(L)\}}{\pi^*(L)}\,\middle|\,L\right] = \frac{\Pr[A=1\mid L]\;\mathrm{E}[Y-b^*(L)\mid A=1,L]}{\pi^*(L)} \qquad \text{(only } A=1 \text{ contributes)}$$
$$= \frac{\pi(L)\{b(L)-b^*(L)\}}{\pi^*(L)} \qquad \text{(definitions of } \pi, b\text{)}$$

Next, identify the counterfactual mean:

$$\begin{aligned}
\mathrm{E}[Y^{a=1}] &= \mathrm{E}[\mathrm{E}[Y^{a=1}\mid L]] && \text{(iterated expectation)}\\
&= \mathrm{E}[\mathrm{E}[Y^{a=1}\mid A=1,L]] && \text{(exchangeability; positivity, so the conditioning is defined)}\\
&= \mathrm{E}[\mathrm{E}[Y\mid A=1,L]] && \text{(consistency)}\\
&= \mathrm{E}[b(L)] && \text{(definition of } b\text{)}
\end{aligned}$$

Each of the three terms $b^*(L)$, $A\{Y-b^*(L)\}/\pi^*(L)$ and $b(L)$ is integrable: $b^*(L)$ by assumption; $A\{Y-b^*(L)\}/\pi^*(L)$ because its absolute value is at most $(|Y|+|b^*(L)|)/\epsilon$; and $b(L)$ because the steps of the identification display hold conditionally on L, giving $b(L)=\mathrm{E}[Y^{a=1}\mid L]$, so $|b(L)|\le \mathrm{E}[|Y^{a=1}|\mid L]$, whose expectation $\mathrm{E}[|Y^{a=1}|]$ is finite. Then:

$$\begin{aligned}
&\mathrm{E}\left[b^*(L)+\frac{A\{Y-b^*(L)\}}{\pi^*(L)}\right]-\mathrm{E}[Y^{a=1}]\\
&= \mathrm{E}\left[b^*(L)+\frac{A\{Y-b^*(L)\}}{\pi^*(L)}\right]-\mathrm{E}[b(L)] && \text{(identification display)}\\
&= \mathrm{E}[b^*(L)]+\mathrm{E}\left[\frac{A\{Y-b^*(L)\}}{\pi^*(L)}\right]-\mathrm{E}[b(L)] && \text{(linearity; each term integrable)}\\
&= \mathrm{E}[b^*(L)]+\mathrm{E}\left[\frac{\pi(L)\{b(L)-b^*(L)\}}{\pi^*(L)}\right]-\mathrm{E}[b(L)] && \text{(iterated expectation and the first display)}\\
&= \mathrm{E}\left[b^*(L)-b(L)+\frac{\pi(L)}{\pi^*(L)}\{b(L)-b^*(L)\}\right] && \text{(linearity)}\\
&= \mathrm{E}\left[\left(\frac{\pi(L)}{\pi^*(L)}-1\right)\{b(L)-b^*(L)\}\right] && \text{(factor out } b(L)-b^*(L)\text{)}\\
&= \mathrm{E}\left[\pi(L)\left(\frac{1}{\pi^*(L)}-\frac{1}{\pi(L)}\right)\{b(L)-b^*(L)\}\right] && \text{(factor out } \pi(L)\text{)}
\end{aligned}$$

If $\pi^*=\pi$, the factor $1/\pi^*(L)-1/\pi(L)$ is 0; if $b^*=b$, the factor $b(L)-b^*(L)$ is 0. Either way the expectation is zero. □

The book (Hernán and Robins 2020, Technical Point 13.2, p. 184) displays this difference with the opposite sign, $(1/\pi(L)-1/\pi^*(L))$; the derivation here gives the sign shown, which does not affect the conclusion.

Second-order bias. This difference depends on the *product* of the errors in $1/\pi$ and in b. This property lets doubly robust estimators that use machine learning for π and b achieve small bias in high-dimensional settings (Chapter 18), where no parametric model is expected to be correct.

Remark 4.3 (From the functional to the estimator). Suppose $\hat{b}$ and $\hat{\pi}$ are predictions from parametric models whose parameter estimates converge in probability, so that $\hat{b}$ and $\hat{\pi}$ settle down to fixed functions b^* and π^*, with π^* bounded away from 0. Then, under the usual regularity conditions, the AIPW estimator of Definition 4.3 converges in probability to the functional in Theorem 4.1, evaluated at these limits. A correctly specified outcome model gives $b^*=b$, and a correctly specified treatment model gives $\pi^*=\pi$; either one makes the difference in Theorem 4.1 vanish. So the AIPW estimator is consistent for $\mathrm{E}[Y^{a=1}]$ whenever at least one of the two models is correct: it is doubly robust in the sense of Definition 4.2.

ℹ Technical Point 13.3: The Augmented IP Weighted Estimator and the Doubly Robust Plug-In Estimator

This technical point shows that the doubly robust plug-in estimator of Fine Point 13.2 is the AIPW estimator in which the outcome model is fit so that the augmentation term sums to zero in every sample (a TMLE). Using two clever covariates, $A/\hat{\pi}(L)$ and $(1-A)/\{1-\hat{\pi}(L)\}$, also makes each plug-in counterfactual mean doubly robust.

5 13.5 How Seriously Do We Take Our Estimates? (pp. 181-185)

Example 5.1 (NHEFS: First Estimates from Real Data)**.** Our first estimates from real data, by IP weighting and by the parametric g-formula:

- mean weight gain: 5.2 kg if everybody had quit, 1.7 kg if nobody had quit
- effect: **3.5 kg** by both methods (95% CI: 2.5, 4.5 for IP weighting; 2.6, 4.5 for the g-formula)

Agreement across methods (and, in the next chapters, g-estimation, outcome regression, and propensity scores) is reassuring because each relies on different modeling assumptions. But observational estimates remain open to serious criticism.

5.1 Three Groups of Conditions

Remark 5.1 (Conditions for a causal interpretation).

1. **Identifiability**: exchangeability, positivity, and consistency, so that the observational study emulates the target trial.
2. **No measurement error** in A, Y, or L (Chapter 9).
3. **No model misspecification** (Chapter 11).

- **Exchangeability**: quitters and non-quitters must be exchangeable given the 9 measured covariates; unmeasured confounding (Chapter 7) or selection bias (Chapter 8) would violate it.
- **Positivity**: the distribution of L in quitters must fully overlap that in non-quitters. Outcome-regression methods (including doubly robust ones) can proceed without positivity if one trusts the outcome model to extrapolate (Fine Point 13.1).
- **Consistency**: there are multiple versions of quitting (gradually, abruptly) and of not quitting (smoking more, smoking less). The estimate corresponds to a vague intervention assigning these versions with their observed frequencies; other interventions could give different effects, which matters for policy and clinical decisions (Hernán and Robins 2020, 182).
- **Measurement**: some mismeasurement of most variables is unavoidable.
- **Models**: e.g., if the true functional form of age in the treatment model is a complex polynomial rather than a parabola, IP weighting will not fully adjust for confounding even if all confounders are measured. Model misspecification acts much like measurement error in the confounders.

5.2 Sensitivity Analysis and Skepticism

> ⚠ The Conditions Are Not Testable
>
> - None of these conditions is empirically testable; we assume they hold approximately, based on **expert knowledge**.
> - Expert knowledge is incomplete, so analyses should explore **sensitivity** to the assumptions.

Sensitivity analyses discussed in the book include those for confounding (Fine Point 7.1; negative outcome controls, Technical Point 7.5; g-estimation, Fine Point 14.2), selection bias (Fine Point 12.1), and model misspecification (Section 11.5), as well as quantitative bias analysis (Fine Point 10.2). Alternative unverifiable conditions, such as those for instrumental variable estimation (Chapter 16), proximal causal inference (Technical Point 7.3), or the front door criterion (Technical Point 7.4), can also be used.

The book's closing message: skepticism of observational causal inferences should be grounded in expert knowledge about each assumption, and "we only take our effect estimates as seriously as we take the conditions that are needed to endow them with a causal interpretation" (Hernán and Robins 2020, chap. 13, p. 183).

6 Summary

Key concepts introduced:

1. **Standardized mean**: $\sum_l \mathrm{E}[Y \mid A = a, C = 0, L = l] \Pr[L = l] = \mathrm{E}[\mathrm{E}[Y \mid A = a, C = 0, L]]$
2. **Parametric g-formula**: estimate the conditional mean with a parametric model and average predictions over the empirical distribution of L (the four-step algorithm)
3. **NHEFS**: standardized means 5.18 kg (quit) and 1.66 kg (did not quit); effect 3.5 kg, essentially the same as IP weighting
4. **Bootstrap**: resample individuals with replacement to estimate the standard error
5. **IP weighting vs standardization**: equal without models; with models they rely on different assumptions, so compare them
6. **Doubly robust estimators** (Definition 4.2): consistent if either the treatment or the outcome model is correct
7. **Conditions for causal interpretation**: identifiability, no measurement error, no model misspecification

Looking ahead:

- Chapter 14 introduces g-estimation of structural nested models, which models the treatment given L together with a model for the causal effect itself.
- Part III defines the g-formula, and the parametric g-formula, for time-varying treatments.
- Chapter 18 returns to doubly robust estimation with machine learning.

Bang, Heejung, and James M Robins. 2005. "Doubly Robust Estimation in Missing Data and Causal Inference Models." *Biometrics* 61: 962–73.

Hernán, Miguel A, and James M Robins. 2020. *Causal Inference: What If*. Chapman & Hall/CRC. https://miguelhernan.org/whatifbook.

Robins, James M. 1986. "A New Approach to Causal Inference in Mortality Studies with a Sustained Exposure Period – Application to Control of the Healthy Worker Survivor Effect." *Mathematical Modelling* 7: 1393–512.